

Research Article

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Method of fundamental solutions with randomized SVD for inverse heat source problem

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Abstract: The study considers the numerical reconstructions of a time-dependent source in a one-dimensional heat source problem. The classical heat equation is used for this reconstruction for the inverse problem. As the inverse problem involves reconstructing boundary and initial conditions and parameter estimation, it is a difficult problem in nature because it is ill-posed. That means small perturbations in data can lead to large errors to the solution. To overcome this difficulty and for numerical implementation, the method of fundamental solutions (MFS) technique is used to avoid meshing of the domain to reduce computational errors. The fundamental solutions are used in the MFS, where boundary conditions, initial conditions, and an additional condition are evaluated, which provides the temperature distribution inside the domain. As a result, a system of linear equations is obtained which is ill-conditioned, which will be in the form of resulting matrix is obtained. To tackle the ill-conditioning of the matrix, the regularization technique is required. The proposed regularization technique is randomized singular value decomposition (rSVD) which approximates the dominant spectral components of the resulting matrix by projecting the matrix onto a low-dimensional randomly generated subspace. This approach significantly reduces computational cost while preserving the essential features needed for stable solution reconstruction. To compare the rSVD approach, the classical Tikhonov regularization is used for evaluating the performance of the rSVD regularization technique.

Keywords: Inverse heat source problem, Method of Fundamental Solutions, Randomized Singular Value Decomposition.

1 Introduction

The parabolic heat partial differential equation describes the study of diffusion processes that arise in real-world applications like porous media flow, thermal engineering and design, financial mathematics, image processing, computer vision, and manufacturing processes. Due to the importance of the heat equation, the study involves both forward problems, which can be referred to as direct formulation, and backward problems, which are named inverse problems.

Inverse heat problems involve the study of reconstruction of initial conditions, boundary conditions, and thermal properties from partial knowledge or measurements of temperature at selected interior thermocouples in the domain [1]. This paper describes the reconstruction of an unknown source term that represents heat generation within the medium.

In this paper, we investigate the reconstruction of an unknown source function depending on time $f(t)$ in the heat equation. In general, full recovery of the source function is not possible from restricted boundary measurements because of limited and noisy measurements. In real cases, the inverse problems are mostly unstable and require significant effort to solve. The major difficulty is the ill-posedness of the problem; for

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that reason, numerical techniques for approximating the solution fail, and the ill-conditioning of the matrix becomes very high [2].

Several numerical methods have been proposed for inverse problems [3–5]. In this paper, we used the linear combination of fundamental solutions to find an approximate solution, and by evaluating points inside the domain, initial conditions, and boundary conditions, we obtain a system of linear equations. This proposed method is called the method of fundamental solutions (MFS) [6–8]. This method is a mesh-free approach for reconstruction of the source term, which is more successful than other numerical techniques because the inverse problem is ill-posed, so it poses challenges for popular numerical techniques like the finite element method (FEM) [12] and the finite difference method (FDM) [13, 14] to perform well for these problems.

The resulting system is ill-conditioned, which means that small perturbations in the data may lead to large errors in the solution [15]. Consequently, direct inversion of the resulting system matrix is numerically unstable, and regularization techniques are required. To mitigate this problem, we used randomized SVD [16, 17]. Randomized SVD aims to efficiently approximate the dominant spectral components of the resulting matrix by projecting the matrix onto a low-dimensional randomly generated subspace. This approach significantly reduces computational cost while preserving the essential features needed for stable source reconstruction. For comparison purposes, we used Tikhonov regularization to check the stability, efficiency, and computational cost by adding 1% and 5% noise to the data [18].

This paper is structured as follows: in section 2, the formulation of the inverse problem is presented; in sections 3 and 4, the MFS framework with required regularization techniques is proposed, and section 5 contains the numerical examples with comparison graphs and tables. Finally, we conclude this research with the outcome of the research.

2 Mathematical formulation

Consider the one-dimensional problem where the source depends on time and satisfies

$$u_t - c^2 u_{xx} = f(t), \quad 0 < x < 1, \quad 0 < t \leq T, \quad (1)$$

$$u(x, 0) = u_0(x), \quad 0 \leq x \leq 1, \quad (2)$$

$$u(0, t) = 0, \quad 0 \leq t \leq T, \quad (3)$$

$$u(1, t) = a(t), \quad 0 \leq t \leq T, \quad (4)$$

with the additional condition that provides the temperature distribution inside the domain to solve the inverse problem

$$u(x', t) = b(t), \quad 0 \leq t \leq T. \quad (5)$$

where $x' \in (0, 1)$ is the interior location of the temperature reading. The functions $u_0(x), a(t), b(t)$ are given for the inverse problem and are consistent at those points where they meet. That means, initial and boundary conditions must not contradict each other at their common points. The problem aims to find the unknown function $u(x, t)$ and specifically recover the unknown source function $f(t)$ by using the above known measurements [19].

We impose an additional assumption on the unknown source because the inverse problem is ill-posed. This additional assumption, in the form of a prior condition, guarantees the unique solvability of the inverse problem [3]. To solve the inverse problem with MFS, by applying the transformation to (1) we converted

the equation into homogeneous form.

$$r(t) = \int_0^t f(\eta) d\eta,$$

$$v(x, t) = u(x, t) - r(t),$$

By using the above transformations, the transformed equation and conditions are given by:

$$v_t - c^2 v_{xx} = 0, \quad 0 < x < 1, \quad 0 < t \leq T, \quad (6)$$

with the initial and boundary conditions

$$v(x, 0) = u_0(x), \quad 0 \leq x \leq 1, \quad (7)$$

$$v(0, t) = -r(t), \quad 0 \leq t \leq T, \quad (8)$$

$$v(1, t) = a(t) - r(t), \quad 0 \leq t \leq T, \quad (9)$$

and the additional condition

$$v(x', t) = b(t) - r(t), \quad 0 \leq t \leq T. \quad (10)$$

By substituting (8) into (9) and (10), finally we obtained the equation

$$v_t - c^2 v_{xx} = 0, \quad 0 < x < 1, \quad 0 < t \leq T, \quad (11)$$

$$v(x, 0) = u_0(x), \quad 0 \leq x \leq 1, \quad (12)$$

$$v(1, t) - v(0, t) = a(t), \quad 0 \leq t \leq T, \quad (13)$$

$$v(x', t) - v(0, t) = b(t), \quad 0 \leq t \leq T. \quad (14)$$

3 Method of fundamental solutions

We employ the Method of Fundamental Solutions with randomized SVD regularization to tackle the ill-posed nature of the problem and enhance convergence. We refer to the fundamental solutions of the one-dimensional heat equation (11) from [20].

$$E(x, t) = \frac{H(t)}{2\alpha(\pi t)^{\frac{1}{2}}} e^{\frac{-x^2}{4\alpha^2 t}}, \quad (15)$$

where $H(t)$ is the Heaviside function. To keep the fundamental solutions non-singular at $t = 0$, the time-shift $\tau > T$ function is introduced to make the resulting MFS system non-singular inside the domain.

$$\phi(x, t) = E(x, t + \tau), \quad (16)$$

is also a non-singular solution of eq (11) in the domain $[0, 1] \times [0, T]$.

Now, let $\Delta = \{(x_j, t_j), j = 1, \dots, n + m + s\}$ be a set of scattered nodes such that

$$\Delta = \Delta_1 \cup \Delta_2 \cup \Delta_3, \quad (17)$$

where

$$\begin{aligned} \Delta_1 &= \{(x_j, t_j), \quad 0 \leq x_j \leq 1, t_j = 0, j = 1, \dots, n\}, \\ \Delta_2 &= \{(x_j, t_j), x_j = 1, \quad 0 < t_j \leq T, j = n + 1, \dots, n + m\}, \\ \Delta_3 &= \{(x_j, t_j), x_j = x_0, \quad 0 < t_j \leq T, j = n + m + 1, \dots, n + m + s\}. \end{aligned}$$

The solution is approximated by using a linear combination of shifted fundamental solutions to approximate the problem equations (11)–(14)

$$v^*(x, t) = \sum_{j=1}^{\infty} \lambda_j \phi_j(x, t), \quad (18)$$

where $\phi_j(x, t) = \phi(x - x_j, t - t_j)$, $\phi(x, t)$ is given by (16) and λ_j are unknown coefficients to be determined. We truncate Eq. (18) by considering a finite number of terms for implementation of the method of fundamental solutions, specifically

$$v(x, t) = \sum_{j=1}^{n+m+s} \lambda_j \phi_j(x, t), \quad (19)$$

For this numerical implementation, we kept the number of source and number of collocation points to be equal, due to which we get unique solution. By evaluating the fundamental solutions at the source and collocation points, we got the linear system of equations, and it will help us to find out unknown coefficients that are given in eq (19)

$$\mathbf{A}\boldsymbol{\lambda} = \mathbf{b}, \quad (20)$$

where $\mathbf{A}$ is a $(n + m + s) \times (n + m + s)$ square matrix:

$$\mathbf{A} = \begin{pmatrix} \phi_j(x_i, t_i) \\ \phi_j(x_k, t_k) - \phi_j(0, t_k) \\ \phi_j(x_l, t_l) - \phi_j(0, t_l) \end{pmatrix},$$

and $\mathbf{b}$ is a $(n + m + s)$ vector:

$$\mathbf{b} = \begin{pmatrix} u_0(x_i) \\ g(t_k) \\ h(t_l) \end{pmatrix},$$

where $i = 1, \dots, n$, $k = n + 1, \dots, n + m$, $l = n + m + 1, \dots, n + m + s$ and $j = 1, \dots, n + m + s$.

In real cases, the temperatures are not measured exactly. There are errors caused by measuring instruments, sensors, environmental effects, numerical errors, and data transmission. Therefore, we don't test our method with exact data, but we do that by adding different levels of artificial noise to the data to check the stability of the inverse problem in Section 5.

4 Regularization Technique

Section 3 gives the linear system of equations for identifying unknown coefficients λ_j . However, the problem is ill-posed. So, the resulting linear system is ill-conditioned. Therefore, solving the system may be highly sensitive to the data. That's why a regularization technique is used to tackle perturbations and noise in the data for a stable approximation of the source function.

Certain regularization methods have been developed to solve these kinds of ill-conditioned systems [21, 22]. We employed randomized SVD (rSVD) [16] to solve the system (3) for correct numerical implementation of the source function. Randomized SVD is a randomized numerical technique used for obtaining a low-rank approximation of a matrix. The rSVD is employed to extract the most useful information from the convolution [23]. It captures the dominant singular values with the help of random projections by identifying a low-dimensional subspace. For ill-conditioned problems, the low-rank approximations are used to diminish the effect of the smaller singular values that are very close to zero and are highly sensitive to noise. Notably, for comparison with the rSVD-based approach, Tikhonov regularization is also considered as a classical regularization technique [18]. The Tikhonov regularization method stands out as one of the oldest and most widely used approaches.

Let $A \in \mathbb{R}^{(n+m+s) \times (n+m+s)}$ be a coefficient matrix of fundamental solutions evaluated at source and collocation points. The singular value decomposition of A is given by

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{(n+m+s) \times (n+m+s)}$ and $V \in \mathbb{R}^{(n+m+s) \times (n+m+s)}$ are orthogonal matrices, and $\Sigma \in \mathbb{R}^{(n+m+s) \times (n+m+s)}$ is a diagonal matrix containing the singular values of A . The matrix Σ contains singular values and provides information about the conditioning of the linear system. Specifically, the small singular values are highly sensitive to perturbations in the data. Therefore, direct computation of SVD is not possible. To reduce the influence of sensitive singular values or unstable components, we used the rSVD approach. The steps of rSVD are given below [24]

1. **Randomized Sampling :** Generate a random Gaussian matrix $\Omega \in \mathbb{R}^{(n+m+s) \times q}$, where $q = k + p$, k is targeted rank and p oversampling parameter. Then, project Ω onto A to get the sample matrix Y .

$$Y = A\Omega,$$

2. **Orthonormalization :** Compute the QR factorization of Y

$$Y = QR,$$

where $Q \in \mathbb{R}^{((n+m+s) \times q)}$ has orthonormal columns. The matrix Q provides the dominant subspace of A .

3. **Project A and Compute smaller SVD :** Project A onto the subspace spanned by Q

$$B = Q^T A,$$

As a result, we get a new and much smaller matrix $B \in \mathbb{R}^{q \times (n+m+s)}$. Now, compute the SVD of the smaller matrix

$$B = \tilde{U}\Sigma V^T,$$

The approximate left singular vectors of A are then obtained as

$$U = Q\tilde{U},$$

Thus,

$$A = U\Sigma V^T,$$

4. **Truncation and solution :** Retain only the first k dominant singular components

$$A_k \approx U_k \Sigma_k V_k^T,$$

The regularized solution will be given by

$$A\lambda = b^\delta,$$

is

$$\lambda_k = V_k \Sigma_k^{-1} U_k^T b^\delta. \quad (21)$$

where k is the retained rank.

To provide a classical benchmark for the proposed rSVD approach, Tikhonov regularization is applied to the system. It is defined as the solution of the following least squares problem

$$\min_{\lambda} \{\|A\lambda - b\|^2 + \alpha^2 \|\lambda\|^2\}, \quad (22)$$

where $\|\cdot\|$ denotes the Euclidean norm and α is called the regularization parameter. In its simplest form, the Tikhonov regularization replaces the system (3) with the regularized system

$$(A^T A + \alpha I)\lambda = A^T b. \quad (23)$$

The selection of parameter α will be made through the generalized cross-validation (GCV) criterion [21].

5 Numerical Results

In this section, numerical examples will be presented so that their performance, accuracy, and efficiency can be analyzed. Three different benchmark examples were processed using the MFS-based approach with both Tikhonov and rSVD regularization techniques, and comparison graphs have been provided by comparing with the exact source.

Additionally, for the purpose of comparison, the thermal diffusivity constant c is assumed to be 1. Measurement errors can be simulated by introducing 1% and 5% noise; stability must be verified before solving the inverse problems. Finally, error graphs are provided to assess accuracy.

Example 1. Let the exact solution for problem (1) – (5) be

$$u(x, t) = e^{-t} \sin(x) + 3tx^2 + \frac{1}{4}x^4, \quad (24)$$

$$f(t) = -6t. \quad (25)$$

We take $x' = 0.5$, and the number of equidistant collocation points is $n = m = s = 11$. The given data $u_0(x)$, $a(t)$, $b(t)$ can be computed from the exact solution.

Figure 1 illustrates the behavior of the singular values of the MFS coefficient matrix A . The singular values decrease rapidly over the first several indices, followed by a very small magnitude for the remaining singular values. This rapid decay indicates that the coefficient matrix is severely ill-conditioned. The condition number of the matrix is $\text{cond}(A) = 5.070329 \times 10^{18}$. Such a large condition number indicates that the MFS linear system is highly sensitive to perturbations in the data.

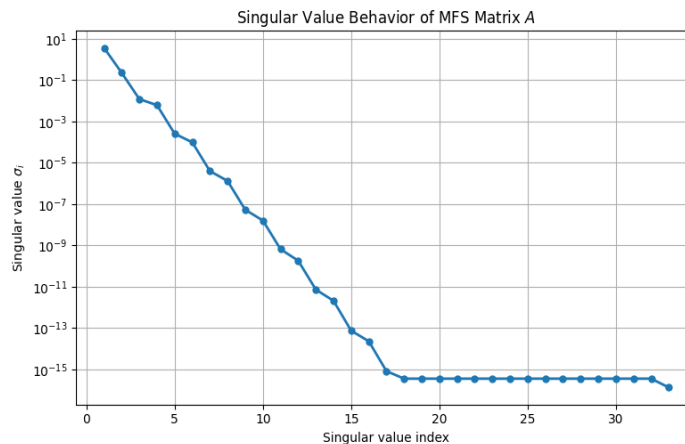


Fig. 1: Singular value distribution of the MFS coefficient matrix A .

The resulting ill-conditioned MFS system is solved using Tikhonov regularization and rSVD. The reconstructed source functions are then compared with the exact source $f(t) = -6t$. To investigate the effect of measurement errors, a relative noise level of 1% is introduced into the following graphs.

The reconstructed heat source obtained using Tikhonov regularization and rSVD is compared with the exact source function in Figure 2. Both regularization approaches provide reconstructions that closely follow the exact source over the considered time interval. The differences between the reconstructed and exact source functions become more visible toward the end of the time interval. The reconstruction errors for both regularization approaches are also examined over the time interval. Figure 3 presents the absolute error between the reconstructed and exact source functions.

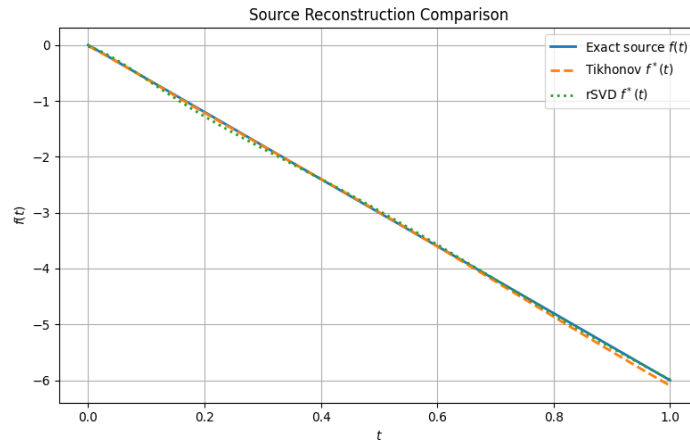


Fig. 2: Source reconstruction with 1% noise, $\tau = 2.5$, $\alpha = 8.191366 \times 10^{-11}$, and $q = 27$.

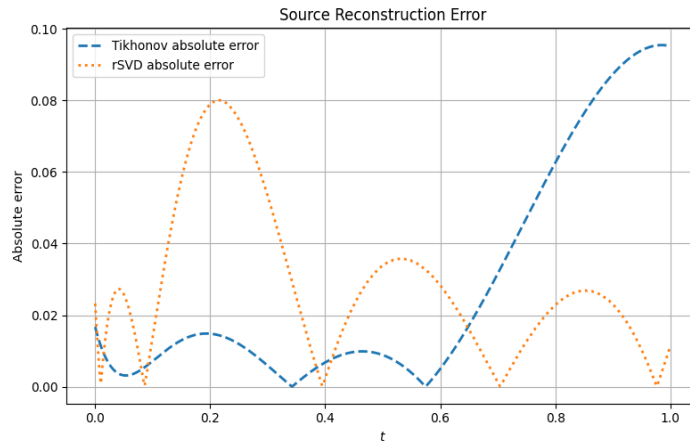


Fig. 3: Absolute reconstruction error with 1% noise.

To examine stability under increased measurement noise, the same reconstruction procedure is applied with a 5% noise level. The reconstructed source and corresponding error plots are compared with the exact source to assess the robustness of both regularization methods against higher data perturbations.

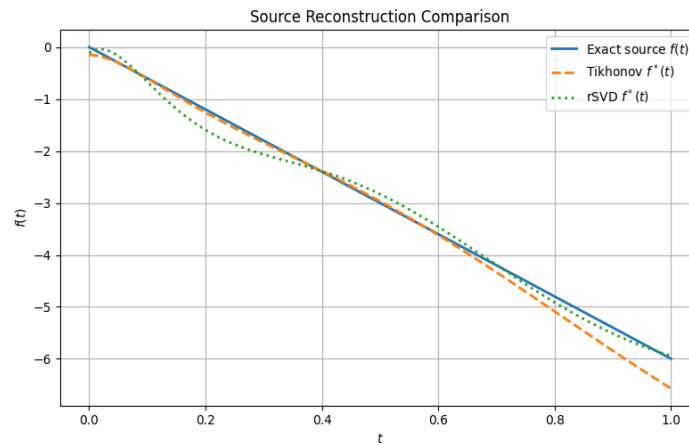


Fig. 4: Source reconstruction with 5% noise, $\tau = 2.5$, $\alpha = 8.191366 \times 10^{-11}$, and $q = 27$.

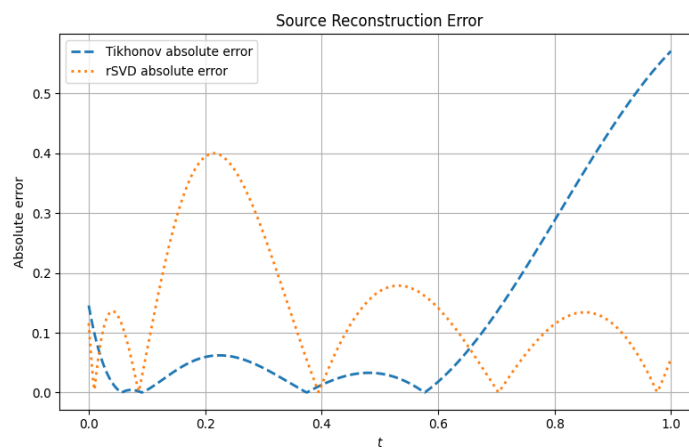


Fig. 5: Absolute reconstruction with 5% noise.

Example 2. To explore how the proposed method can be applied to the inverse heat source problem in greater depth, we look at a more complicated example:

$$u(x, t) = x^2 + 2t + \sin(2\pi t), \quad (26)$$

$$f(t) = 2\pi \cos(2\pi t). \quad (27)$$

Note that in the case of Example 2, $u(0, t)$ is not zero. However, we can define $u = u_1 + u_2$, where u_1 satisfies (1)–(5) with modified data

$$u_1(0, t) = 0, \quad u_1(x', t) = b(t) - u_2(x', t).$$

The second component u_2 satisfies

$$u_{2t} - u_{2xx} = 0, \quad 0 < x < 1, \quad 0 < t \leq 1, \quad (28)$$

$$u_2(x, 0) = 0, \quad 0 \leq x \leq 1, \quad (29)$$

$$u_2(0, t) = 2t + \sin(2\pi t), \quad 0 \leq t \leq 1, \quad (30)$$

$$u_2(1, t) = 0, \quad 0 \leq t \leq 1. \quad (31)$$

Thus, we can first solve problems (28)–(31) as a well-posed problem by MFS, and obtain $u_2(x', t)$, then we can immediately use the proposed technique to solve problem (1)–(5).

We take $x' = 0.5$, and the number of equidistant collocation points is $n = m = s = 11$. The given data $u_0(x)$, $a(t)$, $b(t)$ can be computed from the exact solution.

The singular values of the MFS matrix decay rapidly and then exhibit a small singular-value tail, indicating that the matrix is highly ill-conditioned with condition number $\text{cond}(A) = 2.338976 \times 10^{18}$. This makes the solution extremely sensitive to perturbations in the data. Hence, it is necessary to apply regularization in order to obtain a stable approximation of the source.

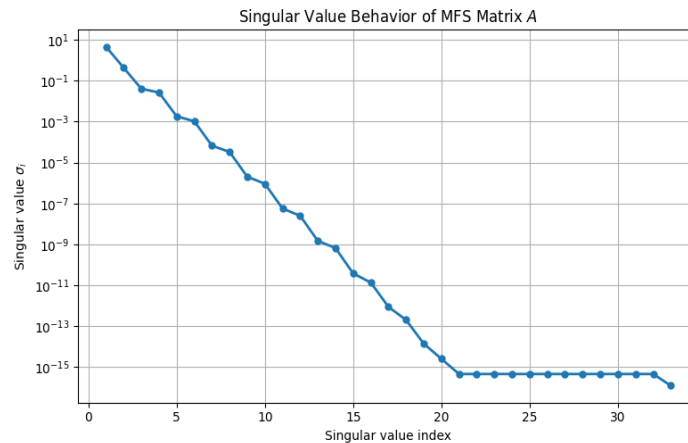


Fig. 6: Singular value distribution of the MFS coefficient matrix A .

The ill-posed MFS system is considered with both Tikhonov and rSVD regularization. For Example 2, we compare the resulting source with the exact function $f(t) = 2\pi \cos(2\pi t)$ under 1% noisy data. The corresponding source reconstructions are shown in figure 7 and their absolute errors measured over the time interval are presented in figure 8. The graphs visualize the reconstruction behavior of the two regularization approaches.

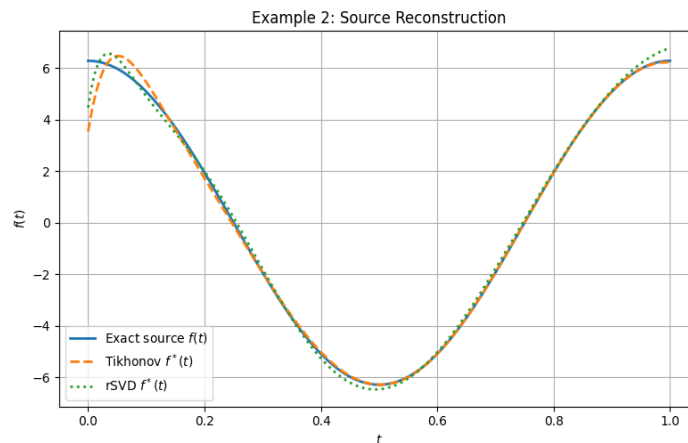


Fig. 7: Source reconstruction with 1% noise, $\tau = 1.7$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 22$.

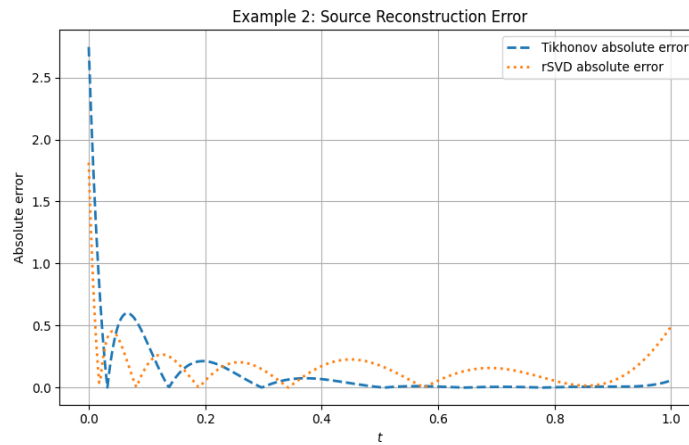


Fig. 8: Absolute reconstruction errors with 1% noise, $\tau = 1.7$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 22$.

The 5% noise results show the reconstructed heat source and the corresponding absolute errors, illustrating the performance of the regularization methods under increased measurement noise.

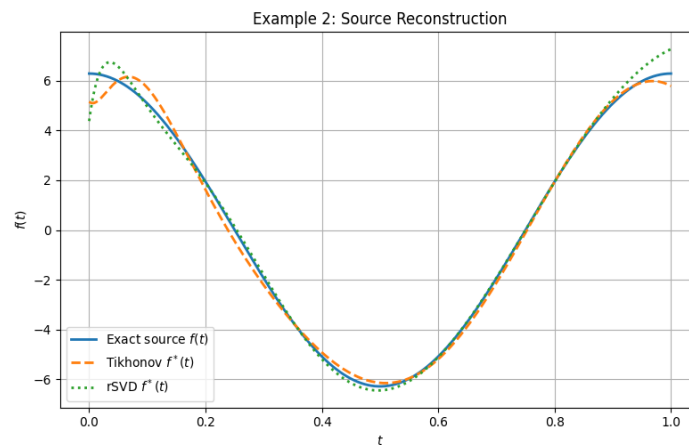


Fig. 9: Source reconstruction with 5% noise, $\tau = 1.7$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 22$.

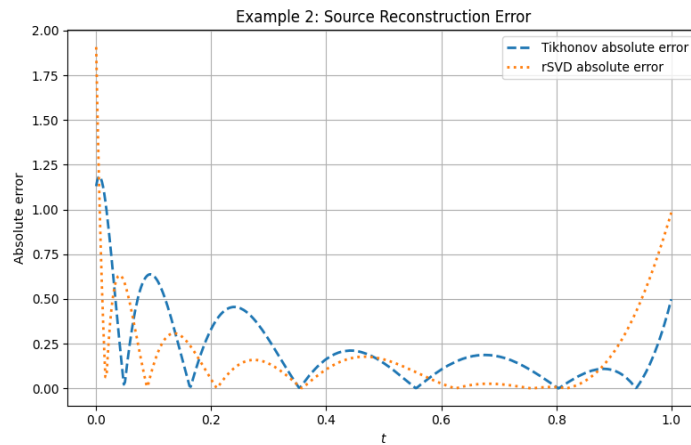


Fig. 10: Absolute reconstruction errors with 5% noise, $\tau = 1.7$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 22$.

Example 3. We consider the following direct problem:

$$u_t = u_{xx} + f(t), \quad 0 < x < 1, \quad 0 < t \leq 1, \quad (32)$$

$$u(x, 0) = 0, \quad 0 \leq x \leq 1, \quad (33)$$

$$u(0, t) = 0, \quad 0 \leq t \leq 1, \quad (34)$$

$$u(1, t) = 0, \quad 0 \leq t \leq 1, \quad (35)$$

with the heat source function

$$f(t) = \begin{cases} -1, & t \in [0, 0.25), \\ 1, & t \in [0.25, 0.5), \\ -1, & t \in [0.5, 0.75), \\ 1, & t \in [0.75, 1]. \end{cases} \quad (36)$$

The singular values decay rapidly, like the previous examples, showing the MFS system to be severely ill-conditioned. The condition number is $\text{cond}(A) = 5.070329 \times 10^{18}$, which confirms this and shows that regularization is required.

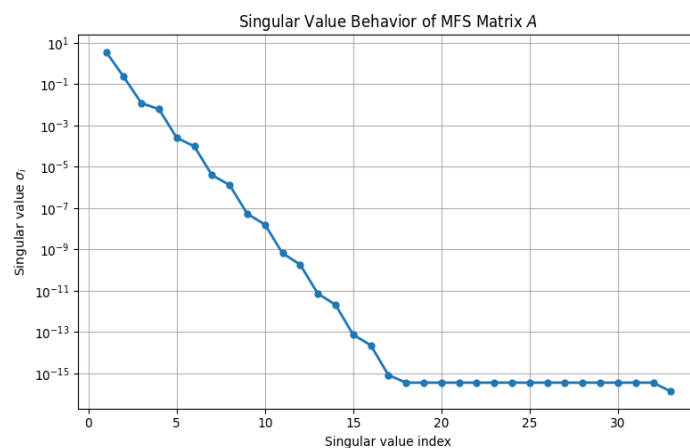


Fig. 11: Singular value distribution of the MFS coefficient matrix A .

For the 1% noisy data, we reconstruct the unknown heat source by using Tikhonov regularization and rSVD. The corresponding source reconstructions are depicted in Figure 12, whereas the absolute reconstruction errors are illustrated in Figure 13. Both figures give an impression of the reconstruction qualities of the regularizations.

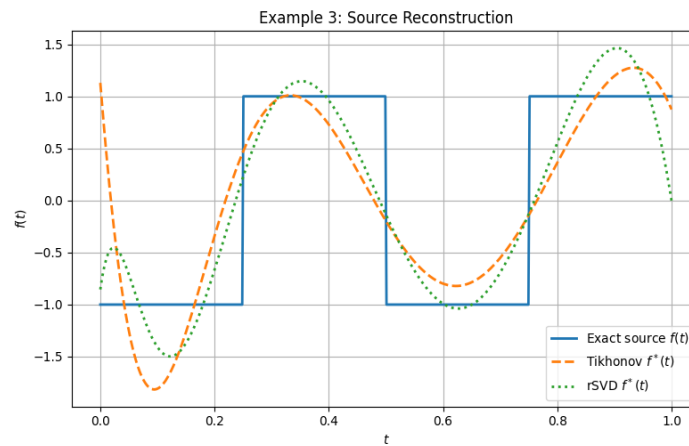


Fig. 12: Source reconstruction with 1% noise, $\tau = 2.5$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 32$.

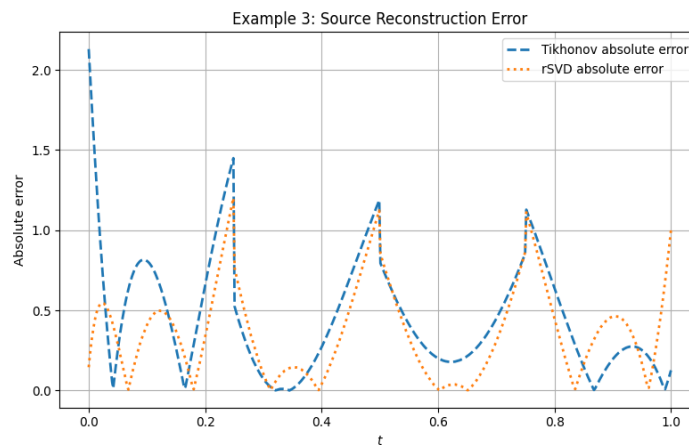


Fig. 13: Absolute reconstruction errors with 1% noise, $\tau = 2.5$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 32$.

For the 5% noisy data, the source reconstructions and corresponding absolute errors are presented in Figures 14 and 15, respectively.

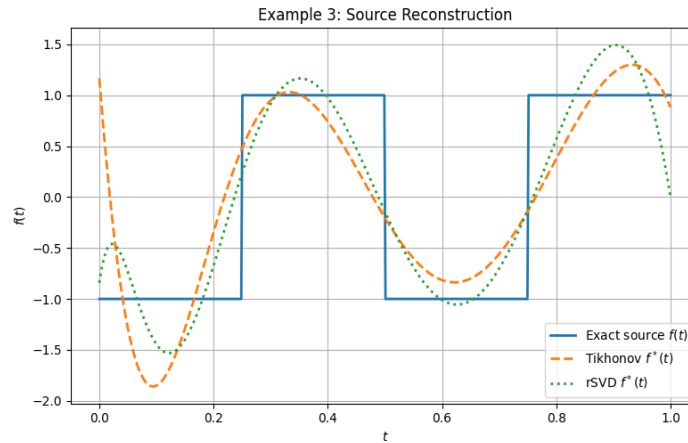


Fig. 14: Source reconstruction with 5% noise, $\tau = 2.5$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 32$.

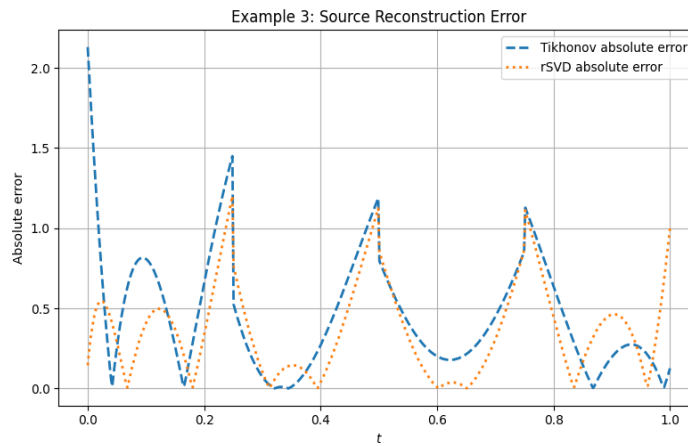


Fig. 15: Absolute reconstruction errors with 1% noise, $\tau = 2.5$, $\alpha = 1.000011 \times 10^{-15}$, and $q = 32$.

6 Conclusion

In this paper, the one-dimensional inverse heat source problem is considered. The classical heat equation is highly used in industry, imaging, and in areas of computer science. So, direct formulation as well as inverse formulation of the heat problem is important for real-world applications. The inverse problem in this paper presented the reconstruction of an unknown source function that depends on time, by keeping the solutions, initial, and boundary conditions in our study.

The inverse heat source problem was ill-posed. So, it requires challenges to solve because the resulting matrix is highly ill-conditioned. The MFS approach is used to tackle the problem numerically, where fundamental solutions of the heat equation are used for reconstruction. In this way, the system of linear equations is obtained. Now, to tackle the ill-posedness, the rSVD regularization is used, which aims to efficiently approximate the dominant spectral components of A by projecting the matrix onto a low-dimensional randomly generated subspace. To facilitate comparison, the Tikhonov regularization is used.

Furthermore, three benchmark examples are solved. For each example, we see that both regularizations have performed well initially when there is 1% noise. As the noise has increased to 5%, due to severe

ill-conditioning, results deviate from the exact results. However, there is a slight edge to rSVD over Tikhonov regularization because, in rSVD, much more careful information is carried by the dominant singular values. That's why, rSVD has performed well in terms of accuracy, stability and also in time efficiency because the python code of rSVD was taking less time than of Tikhonov. Therefore, rSVD provides a computational framework for inverse problems.

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